Fiscal Year:	FY 2018	**Task Last Updated:**	FY 12/24/2018
PI Name:	Nielsen, Sheila Ph.D.		
Project Title:	Genotypic and Phenotypic Responses of Candida albicans to Spaceflight		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology		
Space Biology Cross-Element Discipline:	(1) Reproductive Biology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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Zip Code:	59717	**Congressional District:**	1
Comments:	PI name change to Sheila Nielsen in 2014 (formerly Sheila Nielsen-Preiss)--Ed., 1/12/2015		
Project Type:	Flight	**Solicitation / Funding Source:**	2008 Space Biology--Microbial, Plant, Cell Biology: NNH08ZTT003N
Start Date:	04/01/2012	**End Date:**	07/31/2018
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	4
No. of Master's Candidates:	4	**No. of Bachelor's Degrees:**	7
No. of Bachelor's Candidates:	7	**Monitoring Center:**	NASA ARC
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Flight Program:	ISS		
Flight Assignment:	Space-X ; ISS NOTE: Extended to 7/31/2018 per NSSC information (Ed., 10/10/17) NOTE: Extended to 7/31/2017 per NSSC information (Ed., 7/13/16) NOTE: Extended to 7/31/2016 per A. Chu/ARC and NSSC information (Ed., 8/12/15) NOTE: Extended to 8/31/2015 for reflight as Micro-8 experiment, per A. Chu/ARC and NSSC information (Ed., 4/3/14) NOTE: Extended to 8/31/2014 per A. Chu/ARC and NSSC information (Ed., 6/17/13)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX12AH29G		
Performance Goal No.:			

Performance Goal Text:

Task Description:

Micro-8 Experiment (Addendum March 2014 to "Genotypic and phenotypic responses of Candida albicans to spaceflight")
We had the opportunity to fly the Micro-6 payload of C. albicans experiments on SpX CRS-1 (Commercial Resupply Service-1) in October 2012 and many of the experimental results from this study were positive and intriguing. Subsequent to this initial flight experience, we are in a position to repeat several of the experiments to validate the data and provide power to the analyses. In addition, pursuant to the lessons learned, internally by our team and more generally as part of the initial Space Exploration expedition, we feel refining the experimental design and altering the sample handling processes will strengthen the data.

The goal of our ongoing studies is to further examine the responses of C. albicans to microgravity by performing hypothesis-driven research in spaceflight conditions. Our overarching hypothesis is that exposure of C. albicans to microgravity will alter gene expression and morphology, consistent with a potential increase in virulence. Notably, these studies 1) will inform the value of low shear modeled microgravity (LSMMG) for predicting the physiological responses of C. albicans, 2) will further explore and document the phenotypic parameters of C. albicans that are associated with pathogenicity and are altered during exposure to spaceflight, and 3) may predict conserved responses of higher eukaryotes, including humans, to spaceflight conditions. This re-flight opportunity of the C. albicans payload will strengthen and extend the data as well as focus the analysis and interpretation.

(Ed. note 4/8/14: reporting for the Micro-8 addendum will begin in July 2014)

Micro-6 Experiment

(Addendum to: "The fungal response to simulated microgravity, grant NNX09AM79G")

Modeling the microgravity environment, using rotating suspension culture bioreactors, has allowed investigation of eukaryotic cellular responses in a ground-based model and has provided insights into the impact of spaceflight on cellular physiology. Extrapolation of results from modeled systems to that expected in space is useful and informative, but evaluating the impact of actual space exposure strengthens the validity and interpretation of data from both environments. Using the Candida albicans (C. albicans) yeast system, we have demonstrated that simulated microgravity affects differential expression of several groups of genes, including those related to morphogenic transformation, environmental sensing and ergosterol biosynthesis; induces morphologic changes such as filamentation, biofilm formation, and colony morphology; and results in an increased resistance to the antimicrobial agent Amphotericin B. The constellation of these physiological responses is consistent with a potential increase in the virulence of this opportunistic yeast pathogen. The focus of the current proposal is to characterize physiological outcome measures of C. albicans following exposure to flight on the International Space Station (ISS). The overall objective of this proposal is to evaluate the response of C. albicans to spaceflight in order to 1) predict the potential health risk posed by this opportunistic pathogen in the spaceflight environment, and 2) to further characterize the phenotypic and genotypic response of a eukaryotic system to conditions of microgravity.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

The microgravity environment is also one of low fluid shear. Some of our unpublished data suggests that this low fluid shear environment may be contributing significantly to the yeast response. In addition to evaluating the effects of microgravity on yeast growth, morphology, and physiology, we are evaluating the ability of yeast to infect human monocytes. These studies will provide insight into a broader set of parameters affected by spaceflight and provide a more direct analysis of microgravity-induced pathogenicity. Cumulatively, these studies will increase the understanding of the yeast response to microgravity and other environments characterized by low fluid shear, such as in niches within the human body.

Candida albicans (C. albicans) is a commensal organism that often colonizes the human body shortly after birth and over 25% of the general population are carriers [1]. The rate of infection is rising due to a growing number of immunocompromised hosts, long-term antibiotic use, and an increasing presence of medical implants [2, 3]. As an opportunistic pathogen, C. albicans is capable of causing severe, life-threatening illness in immunocompromised hosts. The specific molecular events responsible for converting a normally benign commensal into a highly pathogenic organism, which in systemic infections can produce 30-45 % mortality, have not been fully elucidated and the dichotomy between commensal and pathogenic behaviors of C. albicans is complex [4-6]. Of particular relevance, spaceflight crews represent a potentially at-risk population for infection from this and other opportunistic pathogens based on a compromised immune response they experience during spaceflight [7-9]. An additional concern is that many microbes studied demonstrate characteristics consistent with increased virulence in the low-shear microgravity environment, but more studies are needed to assess risk [10-16].
A variety of phenotypic changes occur in C. albicans, on both the single and multicellular level, enabling it to better infect a host. Low-shear is also an environmental feature of several niches encountered by C. albicans in vivo and can induce phenotypic changes in the organism. Locations such as the periodontal space, skin, vagina, and the brush border of the respiratory and gastrointestinal tracts have been typified as low-shear [17-19]. Therefore, a clearer understanding of the microbial response to conditions of low shear may provide insight into organismal behavior within niches of the human host.

Modeling the microgravity environment, using rotating suspension culture bioreactors, has allowed investigation of eukaryotic cellular responses, of particular interest here is Sacchromyces cerevisiae and C. albicans, in a ground-based model and provided insights into the impact of spaceflight on cellular physiology [20, 21]. Extrapolation of results from modeled systems to that expected in space is useful and informative, but evaluating the impact of actual space exposure strengthens the validity and interpretation of data from both environments. Using the C. albicans yeast, we have demonstrated that low shear modeled microgravity (LSMMG) affects differential expression of several groups of genes, including those related to morphogenic transformation, environmental sensing and ergosterol biosynthesis; induces morphological changes such as filamentation, biofilm formation, and altered colony morphology; and results in an increased resistance to the antimicrobial agent Amphotericin B [22-25]. The constellation of these physiological responses is consistent with a potential increase in virulence in this opportunistic yeast pathogen. The goal of the research funded by this award was to further examine the responses of C. albicans to microgravity by culturing cells on

Task Progress:

the International Space Station. A partnership was developed with BioServe Space Technologies (Boulder, CO) to utilize the Fluid Processing Apparatus (FPA)/Group Activation Pack (GAP) hardware to conduct hypothesis-driven studies in spaceflight conditions, with ground-based controls. The overall objective of this research was to evaluate the response of C. albicans to the spaceflight environment to predict the potential health risk posed by this opportunistic pathogen in the spaceflight environment, and to further characterize the phenotypic and genotypic response of an eukaryotic system to conditions of microgravity. Our overriding hypothesis is that exposure of C. albicans to microgravity alters gene expression and morphology, consistent with a potential increase in virulence. By extension, an underlying goal was to improve the data interpretation and provide context for adaptations observed in LSMMG.

The studies conducted as part of the research project funded herein allowed us to take several steps forward on the aims outlined. Progress on Aim #1 included identifying several gene pathways that were shown to be differentially expressed in spaceflight, the documentation of which is contributing to ongoing and specific analyses. These gene pathways include those responsive to the gas environment, which we suggest is at least partially responsive for the adaptation process, ergosterol biosynthesis, which we believe contributes to antifungal sensitivity, and those genes associated with the filamentation response, which contribute to cell and colony morphology.

We found some different responses between flight and LSMMG in the studies included in Aim #2. Alterations of cell morphology toward a more filamented form were observed in both LSMMG and spaceflight, although the changes in colony morphology are not consistent between the two models of low fluid shear. These differences will be probed by future studies.

Interestingly, we have been able to confirm differential susceptibility to the antifungal agent amphotericin B. There was consistency between LSMMG and spaceflight as it relates to the increased resistance to this antifungal agent. Based on an aggregate of data, including that obtained as part of the other two aims of this project, we have developed a mechanism to explain this increased resistance, namely elevated levels of CO2 cause a disruption in the ergosterol biosynthetic pathway and ultimately reducing the cellular target of the drug. Further interrogation of this mechanism will be continued in future flight and ground-based studies.

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Abstracts for Journals and Proceedings	Preiss K, Nielsen S. "Candida albicans adaptation responses to microgravity." Forum F2.4-0011-18. Presented at Committee on Space Research (COSPAR) 2018 42nd Scientific Assembly, Pasadena, CA, July 14-22, 2018. Committee on Space Research (COSPAR) 2018 42nd Scientific Assembly, Pasadena, CA, July 14-22, 2018. , Jul-2018
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Significant Media Coverage	Wright J. "Doubling down on excellence. Article about PI Nielsen's research." Montana State University online news, June 27, 2018. http://www.montana.edu/news/mountainsandminds/17818/doubling-down-on-excellence ; accessed 12/26/18; 4/10/21., Jun-2018
Significant Media Coverage	Nielsen S. "MSU investigates the effects of pathogenic yeast in astronauts." Montana State University YouTube video, July 23, 2018. https://www.youtube.com/watch?v=Tkxu9XXp0J0 ; accessed 4/10/21., Jul-2018